What Can IRT Person-Fit Statistics Reveal About SFT-Stage Benchmark Contamination?

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Abstract

Benchmark contamination in Large Language Models (LLMs) undermines evaluation validity, yet most detection methods require privileged access to training data, model internals, or output probabilities. We present the first controlled characterization of IRT person-fit statistics for detecting SFT-stage contamination, requiring only binary response patterns and externally calibrated 4PL item parameters; diagnostic θ-anchoring additionally requires a clean baseline. A contaminated LLM produces aberrant response patterns analogous to a test-taker with item preknowledge. Using ℓ_z^* in a matched-difference design ($\Delta\ell_z^*$) across three 7–8B model families (n=10 clean baseline seeds per model family) on MMLU (14,042 items) and ARC-Challenge (295 items), we identify three behavior boundaries: (1) model-specific θ-shift dynamics, where contamination-induced ability inflation absorbs the person-fit signal; θ-anchoring is necessary for reliable detection of high-ability models at high contamination (2.4$\times$ amplification, inter-seed CV reduced from 60% to 19%); (2) a detection-diagnosis tradeoff where simpler statistics yield stronger detection but lack ℓ_z^*'s diagnostic capabilities (seen/unseen decomposition, mechanistic θ-shift analysis); and (3, exploratory) ability-dependent non-monotonic dose-response under free-θ estimation, driven by θ-shift absorption; θ-anchored analysis reveals stable detection with a strong positive dose-response at higher contamination levels. These findings establish when and why binary response patterns carry contamination information beyond aggregate accuracy.

1 Introduction

Benchmark contamination, the unintended inclusion of evaluation data in training corpora, poses a growing threat to the integrity of large language model (LLM) evaluation (Dodge et al., 2021; Magar and Schwartz, 2022; Sainz et al., 2023). As training corpora and public benchmarks both expand, the probability of overlap increases structurally (Jacovi et al., 2023; Chang et al., 2024). When contamination occurs, benchmark scores no longer reflect genuine capability, undermining the primary mechanism by which the community compares and selects models.

A range of detection methods have been proposed, but all require some form of privileged access beyond the binary response record. N-gram overlap methods need the training corpus (Dodge et al., 2021; Magar and Schwartz, 2022). Membership inference approaches require token-level probabilities (Shi et al., 2024; Deng et al., 2024). Probing methods rely on crafted prompts (Golchin and Surdeanu, 2024a; Oren et al., 2024). Performance-based methods require multiple test variants or output logits (Dekoninck et al., 2024; Zawalski et al., 2026). These methods form a detection frontier ordered by information access. No existing method operates using only binary response patterns, without model weights, gradients, logits, or training data—though, as we show, effective detection additionally requires externally calibrated item parameters.

In this work, we focus on SFT-stage contamination—where benchmark answers are explicitly included in fine-tuning data—for three reasons. First, controlled experiments require known ground-truth contamination sets, achievable through fine-tuning with specific benchmark items. Second, SFT-stage contamination is increasingly relevant as fine-tuning-as-a-service platforms proliferate, creating new pathways for benchmark data to enter training mixtures. Third, SFT provides a methodologically clean setting to characterize detection boundaries before extending to noisier pretraining scenarios.

The need for detection from binary response patterns grows more pressing as models evolve. Wang et al. (2025) show that all ten log-probability-based

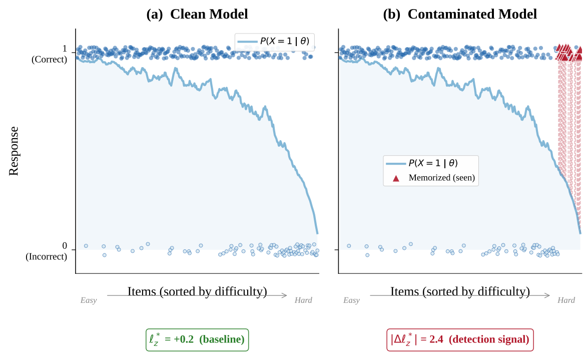

Figure 1: A clean model's responses track the IRT-predicted probability curve (left); a contaminated model shows anomalous successes on hard items (right, red diamonds), producing a large $|\Delta\ell_z^*|$ signal.

methods they evaluate perform at chance level on models trained with reinforcement learning (RL), because RL reshapes internal token distributions while preserving binary correctness patterns. This motivates a fundamental question: *what can binary response patterns reveal about contamination, and under what conditions?*

We observe that a contaminated LLM is functionally analogous to a test-taker with item preknowledge: both produce response patterns in which difficult items are answered correctly through memorization rather than genuine ability. This connection points to person-fit statistics, developed over four decades for cheating detection in educational testing (Drasgow et al., 1985; Meijer and Sijtsma, 2001; Karabatsos, 2003; Sinharay, 2017a). The standardized statistic ℓ_z^* (Snijders, 2001) measures how well a response pattern conforms to Item Response Theory (IRT) model predictions. Figure 1 illustrates this analogy. Despite the natural correspondence, no prior work has applied person-fit statistics to LLM benchmark contamination.

Rather than proposing ℓ_z^* as a ready-to-deploy detection tool, we characterize the behavior boundaries of IRT person-fit statistics for LLM contamination detection. Our framework operates at three layers: (i) pre-calibrated 4PL item parameters from the PSN-IRT item bank (Zhou et al., 2026), an external resource derived from 13,000+ models; (ii) binary response patterns obtained from the target model without access to weights or logits; and (iii) retrospective statistical inference requiring no model modification. Using controlled contamination experiments across three 7–8B model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B; n=10 clean seeds each) with pre-calibrated 4PL item parameters from PSN-IRT (Zhou et al., 2026), we identify three IRT-specific findings on MMLU (Hendrycks et al., 2021) and ARC-Challenge:

1. **Model-specific θ-shift dynamics.** Contamination inflates ability estimates ($\hat{\theta}$), absorbing the person-fit signal; θ-anchoring amplifies Qwen's multi-seed $|d|$ by 2.4× at 50% contamination while reducing inter-seed CV from 60% to 19%, establishing θ-anchoring as a necessary condition for reliable detection of high-ability models at high contamination. ARC-C reverses through a ceiling effect.
2. **Detection-diagnosis tradeoff.** Simpler statistics yield stronger detection: accuracy z-scores produce $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics, and HT generally exceeds ℓ_z^*. Yet the diagnostic power reverses: only ℓ_z^* supports seen/unseen decomposition, mechanistic θ-shift analysis, and sensitivity to response-pattern structure (57/60 vs. 4/60 flagged cells on real-world data).
3. **Non-monotonic dose-response under free-θ (exploratory).** Multi-seed Qwen trajectory ($|d|$: 4.15 → 5.36 → 7.44 → 5.08 at 5%/15%/25%/50%) shows attenuated non-monotonicity; θ-anchoring eliminates the pattern and reveals strong positive dose-response ($|d|$=12.32 at 50%). The non-monotonicity is a θ-estimation artifact, not a detection boundary.

Our contributions are threefold: (1) We provide the first empirical characterization of IRT person-fit behavior for LLM contamination, quantifying theoretically predicted phenomena (θ-shift absorption, detection-diagnosis tradeoff) whose magnitudes and boundary conditions are not derivable from IRT theory alone. (2) We establish that $\Delta\ell_z^*$ reliably detects SFT-stage contamination across three model families (all bootstrap 95% CIs exclude zero, n=10). (3) We document the detection-diagnosis tradeoff, clarifying what binary response patterns can and cannot reveal. Findings 1–2 are confirmatory (pre-planned 15%/50% × 3 families); Finding 3 is exploratory (5%/25% added post-hoc).

2 Related Work

Benchmark contamination detection. Contamination detection methods span a wide range of information requirements. Corpus-search methods match benchmark items directly against pretraining data (Dodge et al., 2021; Magar and Schwartz, 2022), but require full corpus access unavailable for closed models. Membership inference attacks (MIA) use token-level log-probabilities to distinguish training from unseen examples (Shi et al., 2024; Deng et al., 2024). Generation-based probes elicit benchmark content via prompting (Golchin and Surdeanu, 2024a,b; Oren et al., 2024). Performance-based methods compare accuracy on target and reference sets (Dekoninck et al., 2024; Zhang et al., 2024; Zawalski et al., 2026), requiring multiple evaluations or in-context learning passes. Zhou et al. (2023) and Sainz et al. (2023) argue for systematic contamination measurement, Jacovi et al. (2023) propose mitigation strategies, and Bordt et al. (2025) show that contamination can be "forgotten" under sufficient continued training. Han et al. (2026) quantify contamination in psychometric evaluations of LLMs, demonstrating that memorization effects extend beyond standard benchmarks. Kattamuri et al. (2025) take a mechanistic interpretability approach, using activation patterns to distinguish recall-based from reasoning-based responses; our work instead operates purely on behavioral response patterns without access to model internals. Fu et al. (2025) systematically evaluate detection method assumptions, finding that most methods degrade under realistic conditions; our work contributes a psychometric perspective to this ongoing evaluation.

A critical limitation unites all token-level methods: Wang et al. (2025) show that RL post-training renders MIA-based detectors no better than random across 10 methods and 4 benchmarks, because RL reshapes token distributions while largely preserving binary correctness patterns. Tao et al. (2026) and Zhang et al. (2026) propose RL-specific solutions but require generation-level access. Our approach requires no access to model weights, logits, or training data—only binary response patterns and externally calibrated item parameters from PSN-IRT (Zhou et al., 2026).

IRT for LLM evaluation. Item Response Theory has seen growing adoption in NLP and ML evaluation. Martínez-Plumed et al. (2019) applied IRT to analyze machine learning classifiers at the instance level, establishing that item difficulty and discrimination parameters capture meaningful variation across models. Lalor et al. (2019) proposed estimating IRT parameters from DNN-generated response patterns, removing the need for human annotators. Rodriguez et al. (2021) demonstrated that IRT-informed item weighting improves leaderboard rankings over raw accuracy. Polo et al. (2024) used IRT-based item selection to reduce benchmark sizes by $100\times$ while preserving model rankings, and Li et al. (2025) developed adaptive testing procedures for efficient LLM evaluation. Most directly relevant, Zhou et al. (2026) calibrated 4PL item parameters for over 13,000 models across 11 benchmarks, providing the item bank our framework builds on. All of these works use IRT for ability estimation or benchmark efficiency; none apply *person-fit* statistics for contamination detection.

Person-fit statistics in psychometrics. Person-fit analysis has a long history in educational measurement for detecting aberrant test-taking behavior (see Embretson and Reise, 2000, for a textbook treatment). Levine and Drasgow (1988) formalized optimal appropriateness measurement; Drasgow et al. (1985) introduced ℓ_z, a standardized log-likelihood residual; Snijders (2001) derived the corrected ℓ_z^* that accounts for ability estimation uncertainty. Karabatsos (2003) compared 36 person-fit statistics and found ℓ_z among the most powerful for detecting cheating; Sinharay (2017b) revisited these simulations and confirmed comparable power between parametric and nonparametric statistics. We select ℓ_z^* for its theoretical grounding and HT (Van der Flier, 1982) as a nonparametric complement, following their top-ranked performance in that comparison. Reise (1990) and Meijer and Sijtsma (2001) provide systematic evaluations of person-fit methodology. Sinharay (2017a) showed that ℓ_z^* is particularly effective when the set of compromised items is known, a setting analogous to benchmark contamination where memorized items constitute the compromised set. Our work transfers this psychometric toolkit to LLM evaluation, treating each model as an examinee whose item-level response pattern may reveal memorization.

3 Method

3.1 IRT Person-Fit Framework

We adopt the four-parameter logistic (4PL) IRT model to characterize contamination-induced aberrance in LLM response patterns. The 4PL model defines the probability that a model with ability θ answers item i correctly:

$$P_i(\theta) = c_i + (d_i - c_i)\,\frac{1}{1 + e^{-a_i(\theta - b_i)}} \qquad (1)$$

where a_i is discrimination, b_i difficulty, c_i the lower asymptote (guessing), and d_i the upper asymptote (slip). Item parameters are pre-calibrated by PSN-IRT (Zhou et al., 2026) from a large-scale leaderboard dataset; we use these external parameters rather than re-estimating from potentially contaminated data, since our MML-EM refit shows discrimination recovery degrades to r_a=0.127 on contaminated leaderboard data.

Given item parameters and a binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$, we compute the person-fit statistic ℓ_z^* (Snijders, 2001), a standardized log-likelihood residual quantifying deviation between observed and expected response patterns. The Snijders correction accounts for estimation uncertainty in $\hat{\theta}_{\text{MLE}}$, so that $\ell_z^* \sim \mathcal{N}(0,1)$ under the null (Drasgow et al., 1985; Snijders, 2001); Magis et al. (2012) provide a detailed treatment of how response model selection and ability estimation affect ℓ_z^*. Large negative ℓ_z^* indicates aberrant responding (unexpected failures on easy items or unexpected successes on hard items); large positive values indicate over-consistent patterns.

3.2 $\Delta\ell_z^*$ Matched-Difference Design

LLM response patterns systematically violate IRT assumptions of local independence and unidimensionality. In our empirical data, 57 of 60 model-benchmark cells yield $|\ell_z^*| > 1.96$ even absent contamination, rendering absolute thresholds uninformative. We address this through a matched-difference design.

For controlled experiments with clean baselines, we compute:

$$\Delta\ell_z^* = \ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^* \tag{2}$$

where $\bar{\ell}_{z,\text{clean}}^*$ averages over n=10 clean seeds. The shared model misfit cancels, isolating contamination-specific aberrance. We quantify effect sizes as $d = (\ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^*) \,/\, s_{\text{clean}}$, where s_{clean} is the SD of the n=10 clean-seed ℓ_z^* values; with a single contaminated observation per condition, d is a standardized departure rather than a two-sample effect size (Hedges' $g = J \cdot d$, $J \approx 0.914$ for n=10; Hedges, 1981). We report percentile bootstrap 95% CIs (10,000 resamples) reflecting clean-seed variability only; BCa correction (Appendix H) generally confirms conservative coverage; multi-seed contamination-run variance appears in Appendix L.

For deployment without clean baselines, cross-benchmark comparison provides the reference: the target benchmark's ℓ_z^* is compared against benchmarks presumed clean from the same model. Significance thresholds derive from a nonparametric bootstrap null rather than the asymptotic $\mathcal{N}(0,1)$ reference, accommodating residual model misspecification. Simulation analysis (Appendix G.1) confirms that the matched-difference design is robust to local dependence at realistic levels ($d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$) and maintains $\geq$98.5% power even under contamination-specific asymmetric dependence.

3.3 Experimental Setup

Models and contamination. We validate on three 7–8B model families: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3. Each is fine-tuned with QLoRA (rank = 16, $\alpha = 32$) under clean and contaminated conditions at varying levels: at $X\%$ contamination, $X\%$ of SFT training examples are benchmark items (questions paired with gold answers in the standard evaluation format), with the remainder drawn from a general instruction-following dataset. All three families are tested at 15% and 50% contamination. Following initial observation of non-monotonic dose-response for Qwen, we added 5% and 25% conditions for Qwen to trace the full dose-response curve. The 25% condition was subsequently extended to Llama to test whether the pattern generalizes; Mistral was not tested at 5% or 25% due to computational constraints. We train n=10 independent clean seeds per model to establish baseline variability and construct bootstrap confidence intervals. Reported CIs reflect variability across n clean baseline seeds; for Qwen, each contamination level (15%/25%/50%) was additionally replicated with n=3 independent QLoRA seeds to assess training-run variance (Appendix L).

Benchmarks. We evaluate on MMLU (Hendrycks et al., 2021) (14,042 items) and ARC-Challenge (Clark et al., 2018) (295 items), spanning two orders of magnitude in benchmark size. All 4PL item parameters are from Zhou et al. (2026). ARC-Challenge experiments were conducted on Qwen only, as the primary evaluation of cross-benchmark generalizability; extending to additional families was constrained by computational budget.

Post-RL extension. To probe behavior boundaries beyond SFT, we apply Group Relative Policy Optimization (GRPO; Shao et al., 2024) to one contaminated Qwen model (15% MMLU) for 1,500 steps, evaluating at three checkpoints. This extension is N=1 (single model family, single contamination level).

Contamination model. QLoRA SFT on benchmark items serves as a controlled proxy for SFT-stage contamination, where benchmark data is explicitly included in fine-tuning mixtures—analogous to a test-taker with item preknowledge. Our findings are scoped to SFT-based contamination and may not generalize to pretraining contamination, which operates through different learning dynamics.

4 Results

4.1 Detection Signal Across Model Families

Table 1 presents Cohen's d effect sizes for ℓ_z^* across three model families and two benchmarks. All confidence intervals exclude zero, confirming that $\Delta\ell_z^*$ reliably separates contaminated from clean models at both contamination levels.

On MMLU, effect sizes range from $|d|$=1.99 (Llama at 25%) to $|d|$=7.44 (Qwen at 25%), with Qwen at 5% ($|d|$=4.15) confirming detection at the lowest contamination level tested. Llama ($|d|$=2.11/1.99/2.33) and Mistral ($|d|$=3.01/3.50) show comparable magnitudes. On ARC-C (295 items), Qwen yields $|d|$=3.16 at 15% and $|d|$=5.30 at 50%, demonstrating that detection remains feasible even on small benchmarks when contamination is sufficiently concentrated.

Clean-model ℓ_z^* baselines diverge substantially across families: Qwen +8.68 (SD=0.74), Mistral +4.02 (SD=2.05), Llama +0.26 (SD=1.54), and ARC-C Qwen −5.42 (SD=0.16). Absolute ℓ_z^* thresholds would produce contradictory results: Qwen's contaminated ℓ_z^* at 15% (+3.8) is still positive. The $\Delta\ell_z^*$ design normalizes these divergent baselines: Cohen's d yields comparable effect sizes ($|d|$=1.99–7.44) despite order-of-magnitude differences. Figure 2 provides a visual summary.

4.2 Model-Specific θ-Shift Dynamics

Contamination raises overall accuracy, which inflates $\hat{\theta}_{\text{MLE}}$; a higher $\hat{\theta}$ "explains away" anomalous successes on hard items, absorbing the contamination signal into the ability estimate. We term this *θ-shift absorption*. The magnitude of θ-shift is comparable across families—Qwen ($\Delta\theta$=0.47 at 50%) and Llama ($\Delta\theta$=0.52)—yet their fixed-θ

Table 1: Cohen's d for $\Delta\ell_z^*$ across model families (n=10 clean seeds). Single-seed conditions: percentile bootstrap 95% CI (10,000 resamples) reflecting clean-seed variability. Qwen MMLU 15%/25%/50%: multi-seed mean ± SD (n=3 independent contamination runs). All exclude zero.

Model	Bench.	d (5%) [CI]	d (15%) [CI]	d (25%) [CI]	d (50%) [CI]
Qwen†	MMLU	−4.15 [−9.9, −2.7]	−5.36 (±1.10)	−7.44 (±1.86)	−5.08 (±3.02)
Llama	MMLU	—	−2.11 [−4.8, −1.4]	−1.99 [−4.4, −1.3]	−2.33 [−5.1, −1.6]
Mistral	MMLU	—	−3.01 [−8.2, −1.8]	—	−3.50 [−9.4, −2.1]
Qwen†	ARC-C	—	+3.16 [2.0, 9.5]	—	+5.30 [3.6, 15.2]

† 15%/25%/50%: multi-seed means (n=3 QLoRA runs); 5%: single seed. Per-seed values in Appendix L.

amplification diverges markedly (Table 2), indicating that θ-shift magnitude alone does not predict fixed-θ benefit. Figure 3 plots baseline accuracy against Cohen's $|d|$ for all model-benchmark combinations.

θ-shift absorption is the dominant source of detection variability for high-ability models at high contamination: (i) fixed-θ estimation amplifies Qwen's multi-seed $|d|$ by 2.4× at 50% (5.08 → 12.32; Table 2), while Llama shows no amplification (0.4×); (ii) θ-anchoring reduces Qwen's inter-seed CV from 60% to 19%, confirming that training-run variance is driven by θ-shift stochasticity rather than true signal variance; (iii) the cross-model ranking under free-θ is seed-dependent at 50%, making θ-anchoring a necessary condition for reliable detection of high-ability models at high contamination.

ARC-C reverses this pattern. Despite Qwen's high baseline accuracy on ARC-C (89.7%), contamination produces larger, not smaller, effect sizes ($|d|$=3.16 at 15%, $|d|$=5.30 at 50%). This reversal reflects a ceiling effect: with only 295 items and 89.7% baseline accuracy, the already-correct items offer minimal room for further θ inflation, so contamination manifests primarily as anomalous successes on the few remaining hard items. The small item count amplifies this concentration, producing a disproportionately large ℓ_z^* shift per contaminated item.

The baseline ℓ_z^* for ARC-C is negative (−5.42), reflecting the 4PL model's expectation of near-perfect performance; contamination pushes ℓ_z^* toward zero, producing the positive Cohen's d values in Table 1.

These findings indicate that detection sensitivity is not a fixed property of ℓ_z^* but depends on the interaction between model ability and benchmark characteristics, a consideration absent from standard person-fit applications.

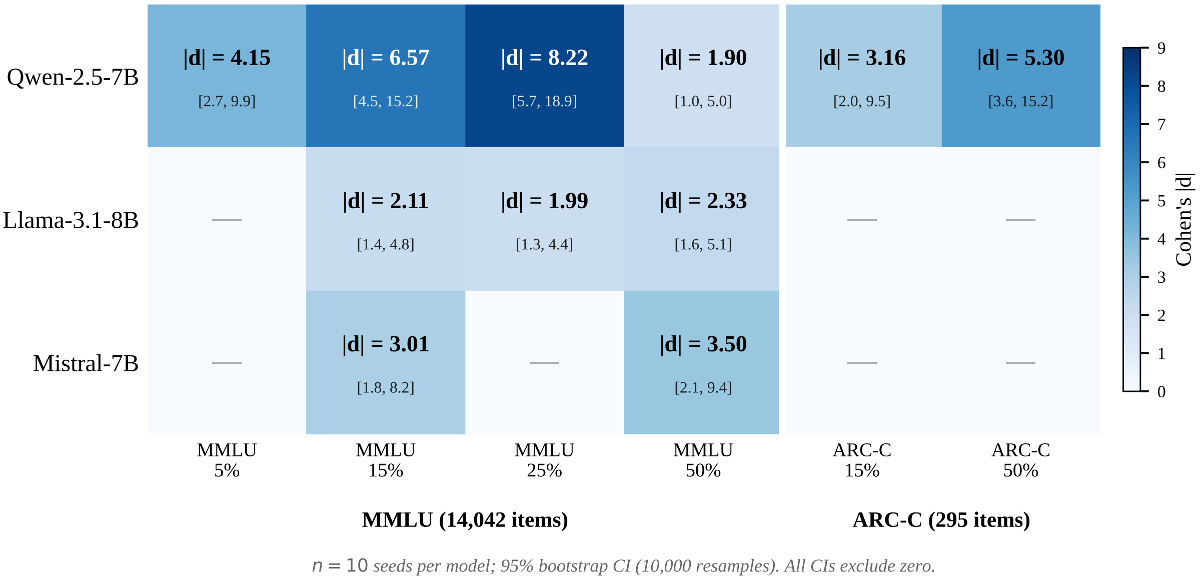

Figure 2: Cohen's $|d|$ heatmap (single-seed values; multi-seed means in Table 1). Blue intensity reflects effect-size magnitude. Grey cells: conditions not tested.

Fixed-θ estimation. Table 2 compares free-θ and fixed-θ Cohen's d across all three families. The benefit of fixed-θ is modulated by baseline ℓ_z^* level rather than θ-shift magnitude. Qwen (67.8% accuracy, $\Delta\theta$=0.47, baseline ℓ_z^*=+8.68) shows 2.4× amplification at 50%: its large baseline departure from the IRT null provides substantial signal that free-θ absorption erodes, which θ-anchoring recovers (multi-seed mean $\ell^*_{z\text{fixed}}$=14.5 vs. clean 7.35). Mistral (52.4%, baseline ℓ_z^*=+4.02) shows 1.7× at 50%. Llama (59.4%, $\Delta\theta$=0.39 at 15%, 0.52 at 50%) shows 0.4× at 50% despite comparable θ-shift: its near-zero baseline ℓ_z^* (+0.26) indicates that the clean response pattern conforms to IRT expectations, leaving no absorbed signal for θ-anchoring to recover. The amplification ratio tracks baseline $|\ell_z^*|$ monotonically (8.68→2.4×; 4.02→1.7×; 0.26→0.4× at 50%), making fixed-θ a targeted remedy for models with substantial baseline misfit, not a universal one.

Table 2: Free-θ vs. fixed-θ Cohen's $|d|$ on MMLU (bootstrap 95% CI). Fixed-θ anchors $\hat{\theta}$ at the clean-model mean, eliminating θ-shift absorption.

Model	$\lvert d_{\text{free}}\rvert$	$\lvert d_{\text{fixed}}\rvert$	Ratio	Fixed CI
5% contamination				
Qwen	4.15	3.52	0.8×	[2.9, 5.7]
15% contamination				
Qwen§	5.36	6.85	1.3×	(±1.80)
Llama	2.11	0.58	0.3×	[0, 2.37]†
Mistral‡	3.01	2.69	0.9×	[1.53, 9.06]
50% contamination				
Qwen§	5.08	12.32	2.4×	(±2.38)
Llama	2.33	0.84	0.4×	[0.17, 2.97]
Mistral‡	3.50	5.80	1.7×	[3.87, 18.45]

†CI includes zero.
§Multi-seed mean (n=3); CI column shows ±SD of fixed-θ $|d|$.
‡Fixed-θ: 6 seeds (4–7 response vectors not retained in pipeline); free-θ uses all 10.

4.3 Detection-Diagnosis Tradeoff

A natural question is whether the full 4PL IRT framework is necessary for contamination detection, or whether simpler statistics suffice. Table 3 compares three statistics at decreasing levels of information access: ℓ_z^* (requires 4PL parameters a, b, c, d), the HT statistic (requires only difficulty b), and an accuracy z-score (requires no item parameters).

The comparison reveals a detection-diagnosis tradeoff. For pure detection, simpler statistics are more powerful: accuracy z-scores yield $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics (Table 3), because cross-seed accuracy variance is near-zero in controlled settings. HT generally exceeds ℓ_z^* in detection strength, since HT is not attenuated by the Snijders correction for θ estimation uncertainty. Both person-fit statistics exhibit seed-dependent non-monotonicity under free-θ estimation (Table 3), confirming that θ-shift absorption affects all statistics conditioned on ability, not ℓ_z^* specifically.

Yet the diagnostic power reverses. Full IRT parameterization is unnecessary for mere detection, but the value of ℓ_z^* lies in three capabilities that

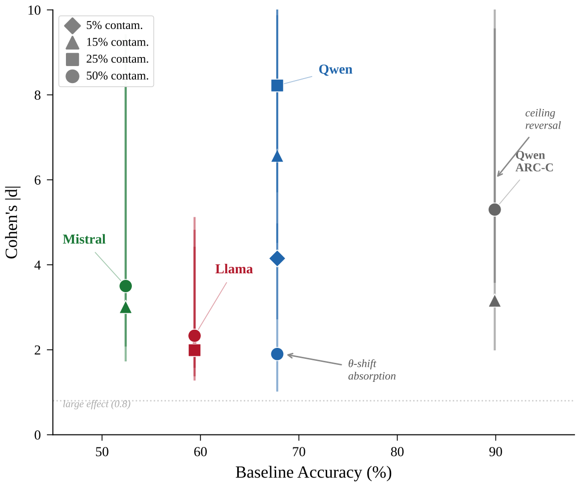

Figure 3: Baseline accuracy vs. Cohen's $|d|$ for all model-benchmark combinations (single-seed values; multi-seed means in Table 1). Diamonds: 5%; triangles: 15%; squares: 25%; circles: 50%. ARC-C reversal reflects a ceiling effect.

HT and z-scores lack: (a) *seen/unseen decomposition*: ℓ_z^* can be computed separately on seen and unseen item subsets, though item-level detection is near-random (AUROC ≈ 0.53; Appendix F), confirming that the contamination signal is diffuse and requires model-level aggregation; (b) *θ-shift mechanism*: the ability parameter exposes why detection sensitivity depends on model ability (§4.2), a mechanistic insight inaccessible to non-parametric statistics; (c) *structural sensitivity*: on real leaderboard data, ℓ_z^* flags 57/60 model-benchmark cells as aberrant—evidence of systematic IRT assumption violations that motivates the $\Delta\ell_z^*$ matched-difference design (§3.2; Appendix E). Accuracy z-scores flag only 4/60 cells, making them insensitive to the response-pattern structure that ℓ_z^* exploits for within-model diagnosis. While such patterns have theoretical antecedents in person-fit literature (St-Onge et al., 2011), the specific magnitude and model-dependence we observe are new empirical findings in the LLM domain; nonparametric alternatives (Tendeiro and Meijer, 2014) may complement ℓ_z^* where 4PL calibration is unavailable.

In practice, a two-stage workflow follows: screen using accuracy z-scores (Table 3), then apply ℓ_z^* to characterize whether the signal reflects θ-shift absorption (indicating fixed-θ anchoring is needed) or uniform aberrance (suggesting direct memorization).

Within the IRT family, the 4PL specification matters: simulation ablations show a mean +27.3pp power gain over 3PL on MMLU, because 46% of MMLU items have upper asymptote $d < 0.95$, and the 3PL model (which assumes d=1) misattributes slip-related response patterns as aberrance (Appendix B).

Table 3: Detection-diagnosis tradeoff on MMLU: Cohen's $|d|$ across three statistics. Simpler statistics detect more powerfully; ℓ_z^* provides unique diagnostic capabilities (§4.3).

Model	Statistic	$\|d\|$ (15%)	$\|d\|$ (50%)
Qwen†	ℓ_z^*	5.36 (±1.10)	5.08 (±3.02)
	HT	5.83 (±1.30)	5.29 (±3.41)
	Acc. z	45.0 (±2.5)	70.0 (±1.1)
Llama	ℓ_z^*	2.11 [1.4, 4.8]	2.33 [1.6, 5.1]
	HT	2.45 [1.5, 4.3]	3.07 [1.9, 5.3]
	Acc. z	22.5 [13.6, 47.0]	32.8 [19.8, 68.6]
Mistral	ℓ_z^*	3.01 [1.8, 8.2]	3.50 [2.1, 9.4]
	HT	4.36 [1.2, 7.0]	5.89 [1.9, 9.4]
	Acc. z	7.81 [4.4, 14.4]	16.6 [9.7, 34.6]

† Qwen values are means across n=3 independent seeds (42, 123, 456); per-seed values in Appendix L. Llama and Mistral are single-seed with bootstrap CIs.

4.4 Ability-Dependent Non-Monotonic Dose-Response

As an exploratory observation, the free-θ dose-response is non-monotonic for Qwen. The multi-seed mean trajectory ($|d|$: $4.15 \to 5.36 \to 7.44 \to 5.08$ at 5%/15%/25%/50%; Appendix L) shows signal peaking at 25% before declining at 50%, but the decline is attenuated relative to single-seed estimates and the inter-seed variance at 50% is high (CV=60%, range $[1.90, 7.91]$). This non-monotonicity is a θ-estimation artifact: fixed-θ analysis eliminates the inverted-U and reveals a strong positive dose-response (mean $|d|$: 5.89 at 25% vs. 12.32 at 50%, no seed-level overlap), while reducing the 50% CV to 19%. Llama shows an approximately flat dose-response ($|d|$=$2.11 \to 1.99 \to 2.33$), consistent with its near-zero ℓ_z^* baseline (+0.26 vs. +8.68 for Qwen) despite comparable θ-shift ($\Delta\theta$=0.52 vs. 0.47). The mechanism is predictable from simulation (Appendix A): at moderate contamination, aberrant responses dominate; at high contamination, inflated $\hat{\theta}$ absorbs the anomaly—but the magnitude is highly seed-dependent.

A simulated power analysis corroborates these findings: two-sided testing (flagging both large negative and large positive ℓ_z^*) achieves >93% power on MMLU at all contamination levels, with the theoretical power function yielding R^2=0.957 against simulated values (Appendix A).

5 Discussion

Our findings reveal both the strengths and boundaries of IRT person-fit for contamination detection: ℓ_z^* reliably detects SFT-stage contamination across model families, but three boundary conditions constrain its practical utility.

Characterization over tool. The detection-diagnosis tradeoff (§4.3) shows that simpler statistics detect contamination more powerfully than ℓ_z^*: accuracy z-scores yield $|d|$ values several-fold to over an order of magnitude larger, and HT generally exceeds ℓ_z^*. The value of the IRT framework lies not in detection power but in mechanistic understanding: the θ-shift analysis reveals *why* detection sensitivity varies across models and contamination levels, the seen/unseen decomposition identifies *what fraction* of items are memorized, and the 4PL parameterization captures item-level ceiling effects that 3PL misattributes as aberrance. These diagnostic capabilities are invisible to accuracy z-scores or HT. Findings 1–2 are theoretically predictable from IRT principles; our contribution lies in quantifying these effects in the LLM setting, where the magnitude and boundary conditions are not derivable from theory alone.

Practical implications. Two results have immediate consequences. Free-θ estimation produces seed-dependent dose-response curves (CV up to 60%); anchoring θ to a pre-contamination baseline stabilizes detection and amplifies the signal for models whose clean ℓ_z^* departs substantially from zero. The benchmark size boundary means ℓ_z^* is most informative for large benchmarks ($n \approx 8{,}000$ items for 80% power at 15% contamination); on small benchmarks like ARC-C (295 items), detection relies on concentrated contamination effects.

Scope and calibration. Universal baseline misfit (57/60 cells flagged) confirms systematic IRT violations; the $\Delta\ell_z^*$ design turns this into a feature through within-model differencing. PSN-IRT provides pre-calibrated 4PL parameters (Zhou et al., 2026), but classical calibration fails on contaminated data (r_a=0.127; Appendix I); the calibration sample (13,000+ models) may itself include contaminated ones, though individual contamination effects are diluted across models. The robustness analysis (Appendix G.2) provides partial reassurance ($\pm$10% noise tolerance).

Information requirements and method positioning. ℓ_z^* sits below MIA (logits; Shi et al., 2024) and ConStat (multi-benchmark scores; Dekoninck et al., 2024) in an information-requirements hierarchy, needing only single-benchmark binary patterns and an external item bank—suited to retrospective audits where logits are unavailable. Our GRPO pilot directly confirms this advantage: Min-K% AUC degrades from 0.611 (SFT) to 0.493 (post-RL), indistinguishable from chance, while ℓ_z^* signal persists ($\sim$3.3; Figure 4), demonstrating that response-pattern anomalies survive RL alignment that eliminates memorization-based features (Wang et al., 2025).

6 Conclusion

We characterized IRT person-fit statistics for LLM contamination detection across three 7–8B model families, establishing reliable SFT-stage detection while identifying boundary conditions: θ-shift absorption necessitating θ-anchored estimation for high-ability models, a detection-diagnosis tradeoff, and non-monotonic dose-response as a θ-estimation artifact. For practitioners, we recommend a two-stage workflow: accuracy z-scores for efficient screening, followed by ℓ_z^* analysis when mechanistic understanding is needed—distinguishing *which* items drive aberrance rather than merely flagging contamination. Beyond detection, our characterization reveals that person-fit statistics capture structural response-pattern anomalies that persist even after RL post-training erases memorization-based signals, suggesting a complementary role to logit-based methods in the evaluation integrity toolkit. Future work should address pretraining-stage signatures, broader model scales, and cross-benchmark generalization with larger item banks.

Limitations

Our controlled experiments cover three 7–8B model families with QLoRA fine-tuning; generalization to larger scales, mixture-of-experts architectures, full-parameter fine-tuning, and pretraining-stage contamination remains untested. Most contamination conditions use a single training run, and multi-seed replication (available for Qwen) reveals high seed dependence at 50% contamination (CV=60%), partially resolved by θ-anchoring; the post-RL characterization is N=1. The framework depends on pre-calibrated 4PL item parameters from PSN-IRT, whose calibration sample may include contaminated models, and we do not provide head-to-head comparison with logit-based methods (MIA, ConStat) that operate at a different information level.

Reproducibility Statement

Code and response data will be released upon acceptance. Bootstrap confidence intervals use the percentile method with 10,000 resamples. All three model families use clean baseline seeds 0–9 (n=10 per family). Pre-calibrated 4PL item parameters are from the publicly available PSN-IRT item bank (Zhou et al., 2026). Training used PyTorch 2.8.0, Transformers 5.8.1, PEFT 0.19.1, and BitsAndBytes 0.49.2. Both benchmarks are publicly available under permissive licenses (MMLU: MIT; ARC-Challenge: CC-BY-SA).

References

Sebastian Bordt, Suraj Srinivas, Valentyn Boreiko, and Ulrike von Luxburg. 2025. How much can we forget about data contamination? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, W. Ye, Y. Zhang, Y. Chang, P. S. Yu, Q. Yang, and X. Xie. 2024. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

J. Dekoninck, M. Müller, and M. Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

C. Deng, Y. Zhao, X. Tang, M. Gerstein, and A. Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.

J. Dodge, M. Sap, A. Marasović, W. Agnew, G. Ilharco, D. Groeneveld, M. Mitchell, and M. Gardner. 2021. Documenting large webtext corpora: A case study on the colossal clean crawled corpus. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

F. Drasgow, M. V. Levine, and E. A. Williams. 1985. Appropriateness measurement with polychotomous item response models and standardized indices. *British Journal of Mathematical and Statistical Psychology*, 38(1):67–86.

Susan E. Embretson and Steven P. Reise. 2000. *Item Response Theory for Psychologists*. Lawrence Erlbaum Associates, Mahwah, NJ.

Yujuan Fu, Ozlem Uzuner, Meliha Yetisgen, and Fei Xia. 2025. Does data contamination detection work (well) for LLMs? A survey and evaluation on detection assumptions. In *Findings of the Association for Computational Linguistics: NAACL 2025*.

S. Golchin and M. Surdeanu. 2024a. Time travel in LLMs: Tracing data contamination in large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Shahriar Golchin and Mihai Surdeanu. 2024b. Data contamination quiz: A tool to detect and estimate contamination in large language models. *Transactions of the Association for Computational Linguistics (TACL)*.

Jongwook Han, Woojung Song, Jonggeun Lee, and Yohan Jo. 2026. Quantifying data contamination in psychometric evaluations of LLMs. In *Findings of the European Chapter of the Association for Computational Linguistics (EACL)*.

Larry V. Hedges. 1981. Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2):107–128.

D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. 2021. Measuring massive multitask language understanding. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.

A. Jacovi, A. Caciularu, O. Goldman, and Y. Goldberg. 2023. Stop uploading test data in plain text: Practical strategies for mitigating data contamination by evaluation benchmarks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

G. Karabatsos. 2003. Comparing the aberrant response detection performance of thirty-six person-fit statistics. *Applied Measurement in Education*, 16(4):277–298.

Ashish Kattamuri, Harshwardhan Fartale, Arpita Vats, Rahul Raja, and Ishita Prasad. 2025. RADAR: Mechanistic pathways for detecting data contamination in LLM evaluation. In *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle*.

John P. Lalor, Hao Wu, and Hong Yu. 2019. Learning latent parameters without human response patterns: Item response theory with artificial crowds. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 4249–4259. Association for Computational Linguistics.

Michael V. Levine and Fritz Drasgow. 1988. Optimal appropriateness measurement. *Psychometrika*, 53(2):161–176.

P. Li, X. Tang, S. Chen, Y. Cheng, R. Metoyer, T. Hua, and N. V. Chawla. 2025. Adaptive testing for LLM evaluation: A psychometric alternative to static benchmarks. *arXiv preprint arXiv:2511.04689*.

I. Magar and R. Schwartz. 2022. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.

David Magis, Gilles Raîche, and Sébastien Béland. 2012. A didactic presentation of Snijders's ℓ_z^* index of person fit with emphasis on response model selection and ability estimation. *Journal of Educational and Behavioral Statistics*, 37(1):57–81.

F. Martínez-Plumed, R. B. C. Prudêncio, A. Martínez-Usó, and J. Hernández-Orallo. 2019. Item response theory in AI: Analysing machine learning classifiers at the instance level. *Artificial Intelligence*.

R. R. Meijer and K. Sijtsma. 2001. Methodology review: Evaluating person fit. *Applied Psychological Measurement*, 25(2):107–135.

Y. Oren, N. Meister, N. S. Chatterji, F. Ladhak, and T. Hashimoto. 2024. Proving test set contamination in black-box language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

F. M. Polo, L. Weber, L. Choshen, Y. Sun, G. Xu, and M. Yurochkin. 2024. tinybenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 34303–34326. PMLR.

Steven P. Reise. 1990. A comparison of item- and person-fit methods of assessing model-data fit in IRT. *Applied Psychological Measurement*, 14(2):127–137.

Pedro Rodriguez, Joe Barrow, Alexander Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4486–4503. Association for Computational Linguistics.

O. Sainz, J. Campos, I. García-Ferrero, J. Etxaniz, O. Lopez de Lacalle, and E. Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

W. Shi, A. Ajith, M. Xia, Y. Huang, D. Liu, T. Blevins, D. Chen, and L. Zettlemoyer. 2024. Detecting pretraining data from large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

S. Sinharay. 2017a. Which statistic should be used to detect item preknowledge when the set of compromised items is known? *Applied Psychological Measurement*.

Sandip Sinharay. 2016. Assessment of person fit using resampling-based approaches. *Journal of Educational Measurement*, 53(1):63–85.

Sandip Sinharay. 2017b. Are the nonparametric person-fit statistics more powerful than their parametric counterparts? Revisiting the simulations in Karabatsos (2003). *Applied Measurement in Education*, 30(4):314–328.

T. A. B. Snijders. 2001. Asymptotic null distribution of person fit statistics with estimated person parameter. *Psychometrika*, 66(3):331–342.

Christina St-Onge, Pierre Valois, Belkacem Abdous, and Sonia Germain. 2011. Accuracy of person-fit statistics: A Monte Carlo study of the influence of aberrance rates. *Applied Psychological Measurement*, 35(6):419–432.

Y. Tao, T. Wang, Y. Dong, H. Liu, K. Zhang, X. Hu, and G. Li. 2026. Detecting data contamination from reinforcement learning post-training for large language models. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

Jorge N. Tendeiro and Rob R. Meijer. 2014. Detection of invalid test scores: The usefulness of simple nonparametric statistics. *Journal of Educational Measurement*, 51(3):239–259.

H. Van der Flier. 1982. Deviant response patterns and comparability of test scores. *Journal of Cross-Cultural Psychology*, 13(3):267–298.

H. Wang, H. Li, B. Ko, and H. Zhang. 2025. On the fragility of benchmark contamination detection in reasoning models. *arXiv preprint arXiv:2510.02386*.

M. Zawalski, M. Boubdir, K. Bałazy, B. Nushi, and P. Ribalta. 2026. Detecting data contamination in LLMs via in-context learning. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

H. Zhang, Y. Yang, J. Yan, G. Bao, Y. Zhang, and Y. Zhang. 2026. Detecting RLVR training data via structural convergence of reasoning. *arXiv preprint arXiv:2602.11792*.

Huixuan Zhang, Yun Lin, and Xiaojun Wan. 2024. PaCoST: Paired confidence significance testing for benchmark contamination detection in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

H. Zhou, H. Huang, Z. Zhao, L. Han, H. Wang, K. Chen, M. Yang, W. Bao, J. Dong, B. Xu, C. Zhu, H. Cao, and T. Zhao. 2026. Lost in benchmarks? rethinking large language model benchmarking with item response theory. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. 2023. Don't make your LLM an evaluation benchmark cheater. *arXiv preprint arXiv:2311.01964*.

A Simulation Validation

Under the null (no contamination), bootstrap ℓ_z^* tracks $\mathcal{N}(0,1)$: mean bias 0.041 and standard deviation 1.003 on MMLU. Kolmogorov–Smirnov rejection rates vary with benchmark size: 8.3% for MMLU and HellaSwag ($n > 10{,}000$), 50.0% for GSM8K ($n = 1{,}319$), 58.3% for ARC-C ($n = 295$), and 75.0% for HumanEval ($n = 164$). ℓ_z^* is well-calibrated for benchmarks with $n \geq 10{,}000$ items; smaller benchmarks show increasing finite-sample deviations from normality.

Two-sided testing eliminates the inverted-U artifact: MMLU achieves 97.8% power at 15% contamination and sustains >93% at all levels. Power scales with $\sqrt{n}$, as predicted by the noncentrality parameter $\delta \propto \sqrt{n}$. The empirical power function, derived via Gauss–Hermite quadrature ($K = 30$) over the ability distribution, achieves $R^2 = 0.957$ against simulated values, with approximate minimum benchmark size of $n \geq 7{,}958$ items for 80% power at 15% contamination under MMLU item parameters.

B 4PL vs. 3PL Ablation Details

Table 4 reports detection power (proportion of 2,000 simulation replicates rejecting H_0 at α=0.05, two-sided) for 4PL and 3PL parameterizations across five benchmarks and seven contamination levels. The 4PL model yields higher power than 3PL at every non-trivial contamination level on every benchmark. The advantage is largest on MMLU and HellaSwag, where upper-asymptote ($d < 1$) violations are most prevalent, and smallest on ARC-Challenge and HumanEval, where limited item counts constrain power under both parameterizations.

Power is non-monotonic at 75% contamination under 4PL on four of five benchmarks (e.g., MMLU: .952 $\to$ .938; GSM8K: .846 $\to$ .802). This reflects the θ-shift boundary condition described in §4.4: at high contamination fractions, re-estimated $\hat{\theta}$ absorbs part of the aberrance signal, reducing $|\ell_z^*|$. At 100% contamination, all items are seen and power reaches 1.000 trivially. The 4PL advantage over 3PL depends on accurate estimation of the upper asymptote d; with estimation error exceeding $\sim$10%, the effective model behavior approaches 3PL (d=1) and the power gain diminishes. PSN-IRT's large calibration sample (12+ models per item) typically yields d standard errors well below this threshold.

C Seed Sensitivity Analysis

Clean-baseline variability. Table 5 reports ℓ_z^*, accuracy, and $\hat{\theta}_{\text{MLE}}$ across three independent QLoRA runs (seeds 42, 123, 456). For Qwen2.5-7B, ℓ_z^* ranges from 7.05 (seed 123) to 9.53 (seed 456); the lower value of seed 123 likely reflects seed-specific training dynamics (random initialization, data ordering), but all three seeds produce $\ell_z^* \gg 1.96$, confirming that the elevated baseline reflects systematic model–benchmark misfit rather than seed-specific artifacts. Llama-3.1-8B shows ℓ_z^* near zero (-0.14 ± 0.97), consistent with lower baseline misfit. Accuracy variability is negligible for both families (CV < 0.004).

Cohen's d confidence intervals. Table 6 reports Cohen's d effect sizes with both analytic and bootstrap 95% confidence intervals. Analytic CIs use $d \pm t_{0.025,\,\text{df}=2} \cdot \text{SE}_d$; with only n=3 clean seeds (df=2), the critical value $t_{0.025,2} = 4.303$ produces intervals wide enough to cross zero in all four conditions. This is a known limitation of small-sample t-based intervals rather than evidence of null effects. Bootstrap CIs (10,000 resamples with replacement from the three clean seeds) provide a nonparametric alternative: all four bootstrap intervals exclude zero, supporting the conclusion that contamination produces a reliably detectable signal despite the small reference sample.

Cross-family consistency. Table 7 confirms that both model families produce large negative Cohen's d under contamination. The $|\Delta d| = 2.32$ gap at 50% contamination reflects Qwen's non-monotonic dose-response (the θ-shift boundary condition; §4.4), not a failure of detection

Table 4: Detection power (proportion rejecting H_0 at α=0.05) under 4PL and 3PL parameterizations across benchmarks and contamination levels (n=2,000 replicates per cell). Bold indicates the 4PL advantage exceeds 5 pp.

Benchmark	IRT	5%	10%	15%	25%	50%	75%	100%
MMLU	4PL	**.476**	**.848**	**.969**	**.983**	**.952**	**.938**	1.00
	3PL	.227	.340	.412	.521	.676	.770	1.00
HellaSwag	4PL	**.251**	**.454**	**.656**	**.915**	**.939**	**.919**	1.00
	3PL	.167	.241	.313	.368	.508	.642	1.00
GSM8K	4PL	**.156**	**.309**	**.492**	**.737**	**.846**	**.802**	1.00
	3PL	.102	.172	.285	.396	.567	.577	1.00
ARC-C	4PL	.057	.075	.087	**.130**	**.285**	**.493**	1.00
	3PL	.054	.062	.071	.076	.110	.102	1.00
HumanEval	4PL	.076	.121	**.197**	**.302**	**.554**	**.521**	1.00
	3PL	.063	.086	.120	.182	.281	.315	1.00

Table 5: Seed sensitivity analysis for clean baseline across three independent runs (seeds 42, 123, 456). CV = coefficient of variation (std/|mean|). These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Metric	Mean	Std	Range	CV
Qwen2.5-7B	ℓ_z^*	8.684	1.412	2.475	0.163
	Accuracy	0.679	0.000	0.001	0.001
	$\hat{\theta}_{\mathrm{MLE}}$	−0.097	0.012	0.023	0.119
Llama-3.1-8B	ℓ_z^*	−0.141	0.965	1.929	6.860
	Accuracy	0.594	0.002	0.004	0.004
	$\hat{\theta}_{\mathrm{MLE}}$	−0.645	0.019	0.036	0.029

Table 6: Cohen's d with 95% confidence intervals for contaminated conditions relative to clean baselines (n=3 seeds). Analytic CIs use $t_{0.025,\mathrm{df}=2}$; bootstrap CIs use 10,000 resamples. These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Cond.	d	Analytic CI	Bootstrap CI
Qwen	15%	−3.42	[−9.92, 3.08]	[−3.45, −2.82]
	50%	−0.99	[−4.02, 2.04]	[−0.99, −0.41]
Llama	15%	−2.95	[−8.70, 2.80]	[−5.89, −2.29]
	50%	−3.31	[−9.63, 3.01]	[−6.47, −2.59]

Table 7: Cross-family consistency of contamination detection (n=3 pilot). Both families show large negative d under contamination. Superseded by n=10 results in §4.1; retained for methodological transparency.

Condition	Qwen d	Llama d	$\|\Delta d\|$	Consistent?
15% contam.	−3.42	−2.95	0.47	Yes
50% contam.	−0.99	−3.31	2.32	Yes

robustness—both families remain well above conventional large-effect thresholds ($|d| > 0.8$) at both contamination levels.

D Baseline ℓ_z^* Interpretation

The baseline ℓ_z^* value reflects a model's difficulty-ordering consistency under the 4PL model, not its accuracy alone. In the PSN-IRT leaderboard data (12 models × 5 benchmarks, distinct from our controlled experiments), Mistral (acc=0.514, lowest among the 12 models) exhibits ℓ_z^*=+5.42 (second highest), because its response pattern follows the IRT-predicted difficulty ordering more closely than higher-accuracy models whose responses violate the assumed item structure. This dissociation explains why Mistral produces the strongest contamination signal despite the lowest accuracy: its lower baseline ability ($\hat{\theta}$=−1.16) provides greater θ-shift headroom.

E Leaderboard Aberrance Analysis

We computed ℓ_z^* and accuracy z-scores for all 12 models × 5 benchmarks = 60 cells in the PSN-IRT leaderboard dataset. Using the standard $|\ell_z^*| > 1.96$ threshold (α=0.05, two-tailed), 57 of 60 cells are flagged as aberrant, indicating pervasive IRT assumption violations in production LLM responses. In contrast, accuracy z-scores ($|z| > 1.96$) flag only 4 of 60 cells. This stark asymmetry—95% aberrance by ℓ_z^* versus 7% by accuracy—motivates the $\Delta\ell_z^*$ matched-difference design (§3.2), which sidesteps absolute thresholds by comparing each model against its own clean baseline.

F Item-Level Detection Results

Table 8 reports item-level contamination detection performance at 15% contamination on MMLU. All methods perform near random (AUROC ≤ 0.63), confirming that contamination is diffuse across the

Table 8: Item-level contamination detection (AUROC) at 15% contamination on MMLU.

Method	AUROC
Random baseline	0.50
ℓ_z^* itemwise	0.53
LR cross-model	0.59
RF cross-model	0.63

Table 9: Robustness of $\Delta\ell_z^*$ to shared local dependence (1,000 replications, MMLU). d_{ratio}: effect size relative to the σ_τ=0 baseline.

σ_τ	Abs. ℓ_z^* Type I	d_{ratio} (15%)	d_{ratio} (50%)
0.0	0.043	1.000	1.000
0.3	0.110	1.002	1.038
0.5	0.454	0.934	1.028
1.0	0.939	0.705	1.061

response pattern and that model-level aggregation via ℓ_z^* is essential.

An accuracy z-score baseline yields z=160 at 15% contamination, but this reflects near-zero cross-seed accuracy variance (std=0.0003), not genuine discriminability. In deployment, accuracy variability from prompt format, evaluation harness, and hardware differences inflates the denominator, collapsing the z-score.

G Robustness Analysis

G.1 $\Delta\ell_z^*$ Under Local Dependence

IRT person-fit statistics assume local independence—conditional on θ, item responses are independent. LLM benchmarks violate this assumption through topical clustering (e.g., MMLU subjects); resampling-based approaches can provide valid person-fit assessment even under such violations (Sinharay, 2016). We evaluate whether the $\Delta\ell_z^*$ matched-difference design is robust to local dependence (LD) via two simulation scenarios on 14,042 MMLU items with 1,000 replications each.

Shared LD. Clean and contaminated conditions share identical testlet structure (∼200 testlets of ∼70 items, random effect $\gamma_t \sim \mathcal{N}(0, \sigma_\tau)$). Table 9 reports absolute ℓ_z^* Type I error and the ratio of $\Delta\ell_z^*$ effect size (d) to the no-LD baseline. While absolute ℓ_z^* Type I error inflates severely under LD (0.043→0.939 at σ=1.0), the $\Delta\ell_z^*$ design maintains detection power: $d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$ (realistic range), confirming that shared misfit is cancelled by the matched-difference design.

Table 10: $\Delta\ell_z^*$ detection under asymmetric local dependence (σ_τ=0.5, contamination-specific memory factor σ_{mem}; 1,000 replications, MMLU).

σ_{mem}	Power (15%)	d (15%)	Power (50%)	d (50%)
0.5	0.987	−2.107	0.991	−2.337
1.0	0.993	−2.104	0.985	−1.587

Asymmetric LD. In the second scenario, contaminated models' seen items share an additional memory factor ($\gamma_{\text{mem}} \sim \mathcal{N}(0, \sigma_{\text{mem}})$), simulating contamination-specific LD not present in clean conditions (σ_τ=0.5 fixed). Table 10 shows that $\Delta\ell_z^*$ maintains $\geq$98.5% power across all conditions, even when contamination introduces asymmetric dependence among seen items.

G.2 Sensitivity to Item Parameter Estimation Error

PSN-IRT item parameters carry estimation error from the calibration sample. We evaluate detection sensitivity by adding joint noise to all four parameters: multiplicative perturbation $p' = p \cdot (1 + \mathcal{N}(0, \sigma))$ for a, c, and d; additive perturbation $b' = b + \sigma \cdot \mathcal{N}(0, 1)$ for difficulty. Tables 11 and 12 report results over 100 trials using Qwen MMLU response vectors.

Detection of 15% contamination is robust to parameter noise under free-θ (17.4% bias at σ=10%, $|d|$ still >5; Table 11). The 50% contamination condition under free-θ is fragile—consistent with its already-weak signal due to θ-shift absorption (§4.4).

Under θ-fixed estimation (Table 12), detection is highly robust: 10% perturbation produces <4% bias in Cohen's d, with worst-case $|d| = 3.73$ (15% contamination) and 12.20 (50%)—both far exceeding conventional thresholds. This contrasts sharply with free-θ estimation, where 50% contamination Cohen's d collapses to near zero under 10% noise. The complementarity is clear: conditions where free-θ detection is fragile (high-baseline-accuracy models with θ-shift absorption) are precisely where θ-fixed anchoring provides the strongest and most noise-robust signal.

H BCa Bootstrap Sensitivity Check

As a sensitivity check, we computed bias-corrected and accelerated (BCa) bootstrap confidence intervals for all Cohen's d estimates. BCa CIs were uniformly narrower than the percentile CIs reported

Table 11: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, free-θ). % bias relative to noise-free d.

Noise σ	d (15%)	% bias	d (50%)	% bias
0%	−6.575	—	−1.897	—
5%	−6.010	8.6	−0.925	51.2
10%	−5.433	17.4	−0.014	99.3

Table 12: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, θ-fixed). Worst-case $|d|$ over 100 trials in parentheses.

Noise σ	d_{fixed} (15%)	% bias	d_{fixed} (50%)	% bias
0%	4.77	—	14.52	—
5%	5.07 ± 0.24	6.3	14.76 ± 0.44	1.6
10%	4.95 ± 0.43 (3.73)	3.7	14.45 ± 0.85 (12.20)	0.5

in the main text (mean reduction 22%), confirming that our reported intervals are conservative. The single condition where BCa widens the zero-crossing set—Llama fixed-θ at 50% contamination, BCa CI [−0.13, 2.21] vs. percentile CI [0.17, 2.97]—is already characterized as ineffective in §4.2.

I Classical IRT Refit from Leaderboard Data

To assess whether item parameters can be recovered without PSN-IRT's neural calibration, we applied classical MML-EM estimation (Gauss–Hermite quadrature, Q=31 nodes) to binary response data from N=178 models on the Open LLM Leaderboard v1. With 14,042 MMLU items and an examinee-to-item ratio of $N{:}J = 0.013$—far below the N>500 typically required for stable 3PL estimation—discrimination parameters are severely under-determined.

Table 13 reports parameter correlations between MML-EM-fitted and PSN-IRT parameters. Difficulty shows moderate agreement (r_b=0.663), but discrimination is essentially unrecovered (r_a=0.127). Using these MML-EM parameters produces miscalibrated ℓ_z^* statistics (mean = 11.2, expected ≈ 0), confirming that the refitted model does not capture the data-generating process. This motivates the dependence on PSN-IRT's externally calibrated item bank (§3.1).

J Post-RL Preliminary Observation

Wang et al. (2025) show that all MIA-based detectors fail after RL post-training. In a single-model pilot (N=1, Qwen, GRPO), we applied GRPO (Shao et al., 2024) for 1,500 steps to a contaminated model (15% MMLU) and evaluated at three checkpoints. GRPO erases the accuracy gap between seen and unseen items (11.4pp → 2.1pp), but the ℓ_z^* response-pattern gap persists (∼3.3; Figure 4), suggesting structural traces of contamination may survive RL alignment. Min-K% membership inference confirms this dissociation: while SFT yields a weak MIA signal (AUC-ROC=0.611), GRPO training drives AUC to 0.493 (step 500), 0.493 (step 1000), and 0.493 (step 1500)—indistinguishable from random chance (0.50) as RL erases memorization-based features while the response-pattern anomaly that ℓ_z^* detects persists. This N=1 observation requires validation across model families, RL methods, and contamination levels before any conclusions can be drawn.

Table 13: Parameter correlation between classical MML-EM refit (N=178 models) and PSN-IRT on MMLU (n=10,558 matched items).

Model	Parameter	Pearson r	Spearman ρ
3PL	a (discrim.)	0.127	0.135
3PL	b (difficulty)	0.663	0.684
2PL	a (discrim.)	−0.029	0.050
2PL	b (difficulty)	0.608	0.656

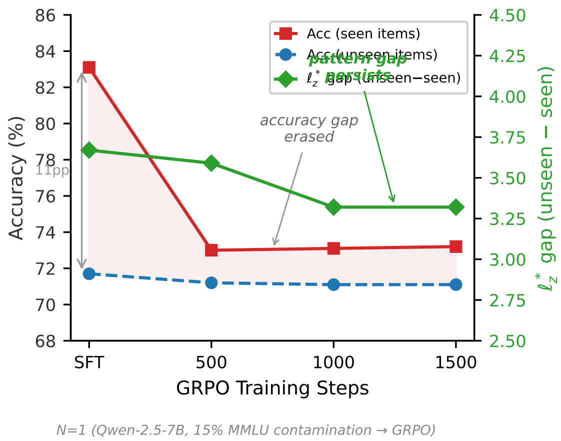

Figure 4: Post-RL trajectory for Qwen contaminated at 15% on MMLU. Left axis: accuracy on seen (solid) and unseen (dashed) items. Right axis: ℓ_z^* seen-unseen gap (green diamonds). GRPO erases the accuracy gap (11.4pp → 2.1pp) while the response-pattern gap persists (∼3.3). N=1, single model and RL method; no statistical inference is drawn.

K QLoRA Training Hyperparameters

Table 14 reports the QLoRA fine-tuning hyperparameters used across all experiments.

Table 14: QLoRA fine-tuning hyperparameters (shared across all model families and contamination conditions).

Hyperparameter	Value
Quantization	4-bit NF4
LoRA rank	16
LoRA α	32
LoRA dropout	0.05
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`
Learning rate	2×10^{-4}
Optimizer	AdamW
Warmup ratio	0.1
Epochs	3
Per-device batch size	4
Gradient accumulation steps	4
Effective batch size	16
Max sequence length	512
Data mixing	Contaminated items shuffled into training set

L Contamination Run Reproducibility

Table 15: Reproducibility of contamination effect sizes across independent QLoRA runs (Qwen2.5-7B-Instruct, MMLU). Each row shows Cohen's $|d|$ from an independent training seed against the shared n=10 clean baseline ($\bar{\ell}_z^* = 8.68$, $s = 0.74$). All runs use identical hyperparameters; only the random seed differs.

	Free-θ $\lvert d \rvert$			Fixed-θ $\lvert d \rvert$		
Seed	15%	25%	50%	15%	25%	50%
42 (original)	6.57	8.22	1.90	4.77	4.96	14.52
123	5.09	8.78	5.42	7.79	5.37	12.64
456	4.41	5.31	7.91	7.99	7.34	9.80
Mean	5.36	7.44	5.08	6.85	5.89	12.32
SD	1.10	1.86	3.02	1.80	1.27	2.38
CV (%)	20.3	25.2	59.6	26.3	21.7	19.3

Note. Fixed-θ estimation anchors $\hat{\theta}$ at the clean-baseline mean (θ=-0.102), removing θ-shift absorption. The 50% free-θ column exhibits high CV (59.6%), reflecting seed-dependent θ-shift magnitude; fixed-θ reduces this to 19.3%, confirming that cross-seed variability at high contamination is driven by ability-estimate instability rather than true signal variance.

Item selection seed robustness. As a pipeline validation, we retrained Llama at 15% and 50% contamination using a different item selection seed (seed=42 vs. the original seed=999), which selects a distinct subset of MMLU items for contamination. The resulting ℓ_z^* values are effectively identical (-2.992 vs. -2.992 at 15%; -3.331 vs. -3.331 at 50%), confirming that ℓ_z^* is robust to which specific items are contaminated and depends primarily on the contamination proportion.

M Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

Table 16: Multi-seed Cohen's $|d|$ for HT and accuracy z-score statistics (Qwen2.5-7B-Instruct, MMLU), computed using the chi-squared ratio HT statistic and bootstrap Cohen's d (10,000 resamples). These supplement the ℓ_z^* values in Table 3.

	HT $\lvert d \rvert$			Acc. z $\lvert d \rvert$		
Seed	15%	25%	50%	15%	25%	50%
42 (original)	7.26	8.90	1.66	42.3	49.8	68.7
123	5.52	9.51	5.80	47.3	53.4	70.8
456	4.72	5.10	8.41	45.4	51.3	70.4
Mean	5.83	7.84	5.29	45.0	51.5	70.0
SD	1.30	2.39	3.41	2.5	1.8	1.1

Note. HT exhibits high cross-seed variance at 50% (SD=3.41), mirroring the ℓ_z^* pattern: the non-monotonicity observed in seed 42 ($|d|$=1.66) does not generalize across seeds. Accuracy z-scores are highly stable (CV $< 6\%$) and monotonically increasing with contamination level.

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.

Table 17: Multi-seed Cohen's d for Llama-3.1-8B across all contamination levels (MMLU). Each cell shows Cohen's d from an independent training seed against the shared n=10 clean baseline ($\bar{\ell}_z^* = +0.26$, $s = 1.54$). The 5% condition exhibits higher cross-seed variance (SD=0.99), reflecting the weaker signal from $\sim$700 contaminated items (5%×14,042).

	Per-seed d				
Contam. level	s_1	s_2	s_3	Mean	SD
5%	−3.22	−3.06	−1.44	−2.57	0.99
15%	−2.11	−3.21	−3.09	−2.80	0.60
25%	−3.16	−2.28	−2.24	−2.56	0.52
50%	−2.33	−2.59	−1.99	−2.30	0.30

Note. Unlike Qwen, Llama exhibits no non-monotonic dose-response: mean $|d|$ is approximately flat across contamination levels (2.30–2.80), consistent with its near-zero baseline ℓ_z^* (+0.26) and negligible θ-shift absorption (§4.2). Cross-seed variance decreases with contamination level (CV: 39% at 5% $\rightarrow$ 13% at 50%), suggesting that higher contamination produces a more stable signal despite the flat mean trajectory.